

CUSTOMER BEHAVIOR ANALYSIS REPORT

**A Data-Driven Study of Shopping Patterns, Revenue Drivers, and Strategic
Customer Insights**

Tools Used: SQL | Python | Power BI
Portfolio Project — Data Analyst

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1. Executive Summary :

This report presents an end-to-end data analytics project examining customer purchasing behavior to identify strategic opportunities for revenue growth. The analysis was conducted using a modern data stack: Python (Pandas) for data wrangling and preprocessing, MySQL for exploratory data analysis (EDA), and Power BI for interactive visualizations. The resulting insights offer a clear roadmap to optimize marketing investments, refine the subscription model, and improve product cross-selling.

2. Dataset Summary :

- Rows: 3,900
- Columns: 18
- Key Features: - Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Methodology & Data Engineering :

A robust data pipeline was established to ensure data integrity prior to analysis:

Data Imputation: Missing customer review ratings were algorithmically imputed using the median rating of their respective product categories.

Feature Engineering: Raw ages were segmented into strategic cohorts (Young-Adult, Adult, Middle-Aged, Seniors). Purchasing string variables (e.g., 'Bi-Weekly', 'Annually') were mapped to quantitative day intervals (14, 365) for time-series evaluation.

Database Integration: All categorical and structural inconsistencies (e.g., spaces and capitalization in column names) were transformed into standardized snake_case format. The refined dataset was then pushed directly to a localized MySQL database via SQLAlchemy for agile querying.

5.Business Transactions Analysis using SQL :

Targeted SQL queries were formulated to answer specific strategic questions. Below are key excerpts from the database environment confirming the insights generated.

Query 1: Revenue Contribution by Gender

This query isolates total revenue grouped by gender, clearly illustrating the significant revenue gap .

```
4  -- Q1. What is the total revenue generated by male vs. female customers?
5
6  • SELECT gender, SUM(purchase_amount) AS total_revenue
7     FROM customer
8     GROUP BY gender ;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
gender	total_revenue			
Male	157890			
Female	75191			

Query 2: High-Value Discount Shoppers

We identified customers who applied discounts yet still spent more than the average purchase amount, representing a segment of highly engaged shoppers.

```
10  -- Q2. Which customers used a discount but still spent more than the average purchase amount?
11  • SELECT
12      customer_id,
13      discount_applied,
14      purchase_amount
15  FROM customer
16  WHERE purchase_amount >
17      (
18          SELECT AVG(purchase_amount)
19          FROM customer
20          WHERE discount_applied = 'YES'
21      );
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
customer_id	discount_applied	purchase_amount		
2	Yes	64		
3	Yes	73		
4	Yes	90		
7	Yes	85		
9	Yes	97		
12	Yes	68		
13	Yes	72		
16	Yes	81		
20	Yes	90		
22	Yes	62		
24	Yes	88		
29	Yes	94		
32	Yes	79		
33	Yes	67		
35	Yes	91		
37	Yes	69		
40	Yes	60		
41	Yes	76		
43	Yes	100		
44	Yes	69		

Query 3: Top Performing Products by Review Rating

Gloves, Sandals, and Boots achieved the highest average satisfaction ratings, highlighting strong product-market fit in these specific items.

```
22
23  -- Q3. Which are the top 5 products with the highest average review rating?
24
25  • SELECT * from
26  (
27      SELECT item_purchased AS products, round(AVG(review_rating),2) AS AVG_review,
28             DENSE_RANK() OVER (ORDER BY AVG(review_rating) desc ) AS ranking
29      from customer
30      GROUP BY item_purchased
31  )t
32  where ranking <= 5 ;
33
```

Result Grid			
Filter Rows:			
Export:  Wrap Cell Content: 			
	products	AVG_review	ranking
▶	Gloves	3.86	1
	Sandals	3.84	2
	Boots	3.82	3
	Hat	3.8	4
	Skirt	3.78	5

Query 4 : Shipping Type Comparison –

Compared average purchase amounts between Standard and Express shipping

```
34  -- Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
35
36  • SELECT * FROM
37  (
38      SELECT shipping_type ,ROUND(AVG(purchase_amount),2) AS AVG_purchasing
39      FROM customer
40      GROUP BY shipping_type
41  )t
42  WHERE shipping_type IN ("Express" , "Standard") ;
```

Result Grid		
Filter Rows:		
Export:  Wrap Cell Content: 		
	shipping_type	AVG_purchasing
▶	Express	60.48
	Standard	58.46

Query 5: Subscription Status vs. Spending Behavior

This query proves the null hypothesis regarding the subscription model: subscribers do not have a statistically higher average spend compared to non-subscribers.

```
44 -- Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.
45
46 • SELECT subscription_status ,COUNT(customer_id) AS total_customer ,
47 ROUND(AVG(purchase_amount),2) AS AVG_spend ,SUM(purchase_amount) AS total_revenue
48 FROM CUSTOMER
49 GROUP BY subscription_status ;
50
51 -- NO , subscribed customers are not spending more.
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	subscription_status	total_customer	AVG_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

Query 6: Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

```
53 -- Q6. Which 5 products have the highest percentage of purchases with discounts applied?
54
55 • SELECT item_purchased,COUNT(*) AS total_purchase,
56 SUM( CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END ) AS discount_purchase,
57 CONCAT(COALESCE(ROUND(SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)*100.0/NULLIF(COUNT(*),0),2),0) ,"%") AS discounted_purchase
58 FROM customer
59 GROUP BY item_purchased
60 ORDER BY discounted_purchase DESC
61 LIMIT 5 ;
62
63
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

	item_purchased	total_purchase	discount_purchase	discounted_purchase
▶	Hat	154	77	50.00%
	Sneakers	145	72	49.66%
	Coat	161	79	49.07%
	Sweater	164	79	48.17%
	Pants	171	81	47.37%

Query 7: Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

```
64  -- Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.
65
66  • WITH customer_type AS (
67      SELECT customer_id,previous_purchases,
68      CASE
69          WHEN previous_purchases=1 THEN "New"
70          WHEN previous_purchases BETWEEN 2 AND 10 THEN "Returning"
71          ELSE "Loyalty"
72      END AS customer_segment
73      FROM customer
74  )
75  SELECT customer_segment ,COUNT(*) AS "No. of customer"
76  FROM customer_type
77  GROUP BY customer_segment;
78
```

customer_segment	No. of customer
Loyalty	3116
Returning	701
New	83

Query 8: Top 3 Products per Category

Listed the most purchased products within each category

```
79
80  -- Q8. What are the top 3 most purchased products within each category?
81
82  • WITH product_counts AS
83  (
84      SELECT category,item_purchased ,
85      COUNT(customer_id) AS total_orders ,
86      ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS product_ranking
87      FROM customer
88      GROUP BY category,item_purchased
89  )
90  SELECT product_ranking,category,item_purchased,total_orders
91  FROM product_counts
92  WHERE product_ranking <=3 ;
```

product_ranking	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Query 9: Repeat Buyers & Subscriptions

Checked whether customers with >5 purchases are more likely to subscribe

```
94
95 -- Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?
96
97 • SELECT subscription_status ,
98     COUNT(customer_id) AS repeat_buyers
99     FROM CUSTOMER
100     WHERE previous_purchases > 5
101     GROUP BY subscription_status ;
102
103
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

Query 10. Revenue by Age Group

Calculated total revenue contribution of each age group.

```
103
104 -- Q10. What is the revenue contribution of each age group?
105
106 • SELECT Age_groups , SUM(purchase_amount) AS total_revenue
107     FROM customer
108     GROUP BY Age_groups
109     ORDER BY total_revenue DESC ;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Age_groups	total_revenue
▶	Young-Adult	62143
	Middle-Aged	59197
	Adult	55978
	Seniors	55763

6. Visual Analytics: Power BI Dashboard :

The centralized Power BI dashboard provides a high-level view of customer distribution, product category performance, and demographic revenue contributions.



- Total customers analyzed: ~3.9K.
- Average purchase amount: \$59.76.
- Average review rating: 3.75/5.
- Clothing is the highest revenue-generating category (~104K).
- Young Adults contribute the highest revenue and sales volume.
- Only 7.3% of customers are subscribers, indicating low subscription adoption.
- Non-subscribers generate significantly higher total revenue than subscribers.

7.BUSINESS INSIGHTS & Recommendations :

```
[40]: df.groupby(["gender", "subscription_status"]).size().reset_index(name="customer_count")
```

```
[40]:
```

	gender	subscription_status	customer_count
0	Female	No	1248
1	Male	No	1599
2	Male	Yes	1053

```
[41]: pd.crosstab(df["category"], df["gender"])
```

```
[41]:
```

gender	Female	Male
category		
Accessories	392	848
Clothing	556	1181
Footwear	199	400
Outerwear	101	223

Female customers contribute purchases across all categories but none of them are enrolled in the subscription program. This indicates a significant untapped growth opportunity and suggests the subscription offering is failing to attract female customers.

Consequently, a critical question emerges: what factors prevent female customers from transitioning from product purchasers to active subscribers?

Recommended Analysis

1. Comparing Female vs Male Spending

```
[42]: #Are females spending less or more ?
df.groupby("gender")["purchase_amount"].agg(
    ["count", "sum", "mean"]
)
```

```
[42]:
```

	count	sum	mean
gender			
Female	1248	75191	60.249199
Male	2652	157890	59.536199

Males are spending more .

2. Identify Female-Dominated Products

```
[45]: # Identify Female-Dominated Products
pd.crosstab(
    df["item_purchased"],
    df["gender"]
)
```

```
[45]:
```

gender	Female	Male
item_purchased		
Backpack	37	106
Belt	55	106
Blouse	66	105
Boots	50	94
Coat	47	114
Dress	52	114
Gloves	37	103
Handbag	58	95
Hat	52	102
Hoodie	51	100
Jacket	54	109
Jeans	29	95
Jewelry	52	119
Pants	48	123
Sandals	59	101
Scarf	45	112
Shirt	59	110
Shoes	48	102
Shorts	48	109
Skirt	49	109
Sneakers	42	103
Socks	58	101
Sunglasses	56	105
Sweater	50	114
T-shirt	46	101

Analysis revealed that several female-oriented products such as Blouses, Skirts, Jewelry, Scarves, Dresses, and Handbags were

predominantly purchased through male customer accounts. This suggests that a significant proportion of purchases may be gift-driven or intended for female family members. The business currently captures the buyer but not the end user, creating a missed opportunity for customer acquisition and subscription growth. A strategic recommendation is to introduce recipient profiling, referral-based subscription programs, family memberships, and targeted female loyalty campaigns to convert end users into direct customers and increase customer lifetime value.

Improvement Ideas :

1. Gift Shopping Program :

Create a "Gift Collection"

Examples:

- Gifts for Her
- Birthday Collection
- Anniversary Collection
- Valentine's Collection
- Mother's Day Collection

Expected impact:

- Easier purchase decisions
- Higher conversion rate
- Larger cart value
- 5–10% increase in average order value

2: Introduce Recipient Profiles :

During checkout:

Is this purchase for:

- Myself
- Wife
- Girlfriend
- Daughter
- Mother
- Friend

Now the company can identify the actual end users which is extremely valuable.

3: Converting Female End Users into Subscribers :

Male buys a handbag => Female uses the handbag.

The company only knows the male. Lost opportunity.

Offer:

"Share this gift and get 20% off her first subscription."

Now:

Male customer → introduces female customer

This is referral-based acquisition.

4: Couple Membership :

Create: Family Fashion Membership

Benefits:

- Shared reward points
- Couple discounts
- Birthday rewards
- Early access sales

This is a strong retention strategy.

6: Female Subscriber Campaign :

Launch:

"Women's Premium Club"

Benefits:

- Free shipping (Express)
- Exclusive launches
- Early access
- Fashion recommendations
- Loyalty rewards
- Free styling recommendations to female influencers

- Clothing dominates both revenue and sales, making it the strongest product category .Increase investment in Clothing inventory and promotions.
- Loyal customers represent the majority of the customer base, creating opportunities for retention programs .So tier-based loyalty programs should be created for repeat customers.
- Launch a targeted campaign for repeat buyers (5+ purchases) who have not yet subscribed .Offer tangible perks: free shipping, early sale access, or cashback to drive subscription sign-ups .Set a realistic goal to grow subscription rate from 7.3% to 15% within 6 months.

```

111 -- product has high ratings but low sales
112 • SELECT
113     item_purchased,
114     COUNT(*) AS total_sales,
115     ROUND(AVG(review_rating),2) AS avg_rating
116 FROM customer
117 GROUP BY item_purchased
118 ORDER BY avg_rating DESC, total_sales ASC
119 LIMIT 5 ;

```

Result Grid			
		Filter Rows:	
		Export:	Wrap
	item_purchased	total_sales	avg_rating
►	Gloves	140	3.86
	Sandals	160	3.84
	Boots	144	3.82
	Hat	154	3.8
	T-shirt	147	3.78

- Promote Gloves, Sandals, and Boots as quality-certified products based on customer review data. Bundle top-rated items with average-rated products to increase cart value and overall

satisfaction scores. Encourage verified reviews on high-volume items (Jewelry, Blouse, Jacket) to build social proof.

- With only 83 new customers, allocate budget to digital acquisition: social ads, search, and referral programs. Offer a compelling first-purchase incentive (e.g., 15% off) to lower the barrier for new customers. Target Young Adults and Middle-Aged segments in acquisition campaigns given their high spend levels.
- Discounts are widely used across products but require profitability monitoring. Optimize discount strategies using product profitability analysis.

CONCLUSION :

This project demonstrates a full-cycle data analytics workflow — from raw data to structured SQL analysis, Python-based exploration, and a polished Power BI dashboard. The analysis revealed a retail business with strong customer loyalty, consistent spending behavior, and a clear category leader in Clothing. Three strategic priorities stand out: converting the large base of unsubscribed repeat buyers, rationalizing discount-heavy products to protect margins, and investing in new customer acquisition to reduce long-term dependency on the existing loyal base. The Power BI dashboard provides business stakeholders with an on-demand, interactive view of customer behavior — enabling faster, more confident decisions across marketing, operations, and product strategy.

